

NATIONAL CYBERSECURITY LAW

Design Project for VIT — Academic Assignment

Subject

Cybersecurity Law &
Policy

Project Type

Group / Individual Project

Total Marks 100

Marks

Deliverables

Report + Presentation + Flowcharts +
Infographics

Important Notice to Students

This is a comprehensive project assignment. Read all sections carefully before starting. Your team must draft a complete Cybersecurity Law for the fictional country of VIT's. Each part of this project corresponds to a section of the law you will create. Be creative, thorough, and technically accurate.

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PART 1: Introduction

Title: The Comprehensive Data Encryption Act (CDEA)

1. Purpose of the Law The Comprehensive Data Encryption Act (CDEA) is enacted to mandate the absolute cryptographic protection of all digital information within VIT's borders. Its primary purpose is to ensure that all personal, corporate, and government data—whether in transit across networks or at rest on local devices and cloud servers—is secured using state-of-the-art encryption methods, rendering unauthorized access useless.

2. Importance of Cybersecurity: The Scenario Context VIT's has achieved unprecedented rapid digital expansion. With a population of over 50 million internet users, the nation's core functions—spanning online banking, e-commerce, cloud-based enterprise, and AI-driven public services—are entirely reliant on digital architecture.

However, this hyper-connectivity has made VIT's highly vulnerable to exploitation. The need for the CDEA is urgent because of a recent surge in severe incidents. The country has faced massive data breaches, ransomware attacks paralyzing hospital infrastructure, and widespread financial fraud. Current perimeter-based security is failing; VIT's needs a law that protects the data itself, ensuring confidentiality even when a system is breached.

3. Goals and Expected Outcomes Upon the enactment of the CDEA, the digital ecosystem of VIT's will shift from a reactive security model to a zero-trust, data-centric defense model. The expected outcomes include:

- **Neutralization of Data Breaches:** Stolen data will be entirely unreadable and useless for extortion or identity theft.
- **Mandatory Standards:** All digital services and devices operating in VIT's must comply with universal cryptographic baseline standards (e.g., AES-256).
- **Restored Public Trust:** Citizens will have absolute confidence that their sensitive financial and personal information is mathematically secured against misuse.

Scenario Context

VIT's is a rapidly growing digital nation with 50 million internet users, online banking, e-commerce platforms, government digital services, smart city infrastructure, AI-based public services, and cloud-based businesses. Recently, VIT's experienced large-scale data breaches, ransomware attacks on hospitals, banking fraud, social media misinformation campaigns, identity theft cases, and cyberbullying incidents.

PART 2: Definitions Section

Define at least 20 important cyber law terms. Your definitions must be precise, legally clear, and relevant to VIT's context. Below are the required terms — you may add more.

Required Terms to Define

#	Term	Definition (To Be Completed by Student)
1	Cybercrime	<i>Any illegal action that are committed using computers or internet network.</i>
2	Hacker	<i>A person which use their technical skills to break into computer system without permission.</i>
3	Ethical Hacker	<i>An authorized expert who test the security of a system to find weakness before bad hackers does.</i>
4	Malware	<i>Malicious softwares designed to cause damage or gain unauthorized access to device.</i>
5	Ransomware	<i>A type of malware that encrypt a victim's files and ask for payment to unlock them.</i>
6	Phishing	<i>When attacker send fake message pretending to be real company to steal sensitive data.</i>
7	Data Breach	<i>An incident where confidential information is copy or stolen by an unauthorized person.</i>
8	Personal Data	<i>Any information that identify a specific person like name, email or phone number.</i>
9	Sensitive Data	<i>Highly private information like bank details or health record which need strong encryption.</i>
10	Digital Signature	<i>A mathematical technique used to prove that a electronic document is authentic and not changed.</i>
11	Electronic Evidence	<i>Any digital data or file which can be present in court to prove a crime.</i>
12	Identity Theft	<i>When someone steal another person personal details to commit fraud in their name.</i>
13	Artificial Intelligence	<i>Computer system that can perform task which usually require human intelligence.</i>
14	Cloud Computing	<i>Storing and accessing data and programs over the internet instead of local computer hard drive.</i>

15	Critical Infrastructure	<i>Important physical or digital systems that a country need to function like power grid or hospitals.</i>
16	Cyberbullying	<i>Using digital platform to harass or threaten someone repeatedly.</i>
17	Deepfake	<i>Fake video or audio created by AI that make it look like someone say or do something they didn't.</i>
18	Encryption	<i>The process to convert readable data into secret code to hide it from unauthorized user.</i>
19	Firewall	<i>A security system that monitor and control incoming and outgoing network traffic based on rules.</i>
20	Zero-Day Exploit	<i>A cyber attack that take advantage of a software vulnerability before the developer know about it.</i>

PART 3: Citizens' Digital Rights

Create a comprehensive Bill of Digital Rights for every citizen of VIT's. For each right, provide a clear explanation of what it means and how it will be enforced.

Required Rights

Right to Privacy Citizens must be protected from unauthorized collection, use, or sharing of their personal data. Any organization are strictly prohibited from collecting user data without clear consent. The personal information must stay confidential and isolated. The National Cyber Security Authority (NCSA) will enforce this by doing random checks, and companies who breaks this rule will face heavy fine.

Right to Secure Data

All organizations must store and protect citizens' data using industry-standard security measures. Under the new encryption act, every company must have to encrypt all citizen data using strong methods like AES-256. This ensure that even if a hacker break in, the data remain unreadable. The IT department of the organization is responsible for this protection and regular audits.

Right to Data Deletion

Citizens may request the complete deletion of their personal data from any organization's systems. A citizen have full right to ask a platform to completely erase their data from the server

and backups. The company must process this deletion request within 15 days. If they fails to delete the data, citizens can report them to the cyber authority for legal action.

Right to Information

Citizens have the right to know what data is collected about them and how it is used. People can demand to see exactly what details a platform keep about them. Organizations has to provide a clear report explaining why they need the data. If any information is wrong, the citizen can force the company to correct it immediately.

Right to Report Cybercrime

Citizens can report cybercrimes without fear of retaliation, and authorities must act within defined timeframes. Every citizen can report cyber attack or fraud without any fear. The government will create a easy online portal for this. The police and NCSA must start investigation within a fixed timeframe after the report is file.

Right to Fair Investigation

Any person accused of a cybercrime is entitled to a fair, transparent, and timely investigation. Any person who is accuse of a digital crime deserve a fair trial. The digital evidence has to be collected transparently without tampering. Everyone get the right to have legal support during the whole process.

Instruction

You must include at least 6 rights. You may add more. Each right should be at least 2-3 paragraphs with specific implementation details.

PART 4: Cyber Offenses and Crimes

Define categories of cybercrime. For each offense, specify a description, real-world example within VIT's context, criminal punishment, and financial fine. Complete the table below.

Crime	Description + Example	Punishment	Fine
Unauthorized Access	Bypassing security to enter a system without permission. Example: Hacker bypass the login page of VIT's city water supply system.	5 years imprisonment	₹5 lakh
Data Theft	Stealing confidential or unencrypted digital files. Example: Hospital patient health records in VIT's was downloaded illegally by an insider.	10 years imprisonment	₹10 lakh
Identity Theft	Pretending to be someone else to commit fraud. Example: Scammer create a fake social media profile of VIT's mayor to ask for money.	5 years imprisonment	₹5 lakh
Phishing	Sending fake emails or links to steal user passwords. Example: Citizens of VIT's received fake banking SMS links that look exactly like real bank.	4 years imprisonment	₹3 lakh
Ransomware	Malware that encrypt files and demand money for the key.	10 years imprisonment	₹20 lakh

	Example: Attackers locked the main database of VIT's central bank and ask for crypto.		
Malware Distribution	Creating and spreading harmful virus softwares online. Example: A fake exam result app spread a dangerous virus across VIT's school networks.	6 years imprisonment	₹8 lakh
Cyberbullying	Harassing or threatening a person on digital platforms. Example: A college student in VIT's was threatened and trolled repeatedly on Instagram.	3 years imprisonment	₹2 lakh
Online Harassment	Sending continuous abusive or unwanted digital messages. Example: A local journalist in VIT's got threatening emails every day from unknown sender.	3 years imprisonment	₹2 lakh
Financial Fraud	Cheating people online for financial gains. Example: Fake UPI payment QR codes was placed in VIT's smart city parking lots.	7 years imprisonment	₹15 lakh

Misinformation	Spreading false news online to create public panic. Example: Fake news about a chemical leak in VIT's caused massive panic on social media.	4 years imprisonment	₹5 lakh
Website Defacement	[Illegally changing the content or look of a website. Example: Hackers change the homepage of VIT's public transport portal with bad images.	5 years imprisonment	₹7 lakh
Cyber Terrorism	Major cyberattack against a nation's critical infrastructure. Example: Terrorist group completely shut down the smart power grid of VIT's capital city.	Life imprisonment	₹50 lakh

PART 5: Data Protection Framework

Create a comprehensive data protection framework that all organizations operating in VIT's must follow. Your framework must cover the full data lifecycle.

Data Collection Rules

Organization should collect only the required data from users. Personal or sensitive data must not be collected without taking user consent first. The data must be collected through a secure channel using HTTPS so it is encrypted during transmission. Companies has to be transparent and tell users exactly why their data is collected.

Data Storage Rules

All the personal and sensitive data should be encrypted and stored using AES-256 encryption algorithm. Regular backup must be done by the company and stored securely in different server. Access to the stored data must be highly restricted. For database administrators, there should be mandatory multi-factor authentication to login.

Data Sharing Rules

Data shouldn't be shared with any third party until user give their explicit consent. Only the necessary information should be shared, not everything. While sharing, the data must always be encrypted using TLS 1.3 or higher protocols to stop interception. Every organization need to maintain a history log of all data sharing activities.

User Consent Requirements

Users should have the ability to withdraw their consent at anytime they want. The consent request form should be written in easy and understandable language, not complex legal words. Consent should be taken for specific task individually. Organization must maintain records of every user's consent in their system securely.

Data Retention Period

When data become expired, it must be securely archived or deleted from the system. The retention schedule must be documented properly and reviewed half yearly by the IT team. Personal data of citizens should be kept only until the main purpose is done. User records must not be kept for more than 5 years after they stop using the service.

Data Deletion Process

Citizens should have the right to request deletion of their data at any time. Organization must review and accept the deletion request within 15 days maximum. All the backup copies of the deleted data must be removed as well. Deleted data must be completely erased using secure data wiping technique so no hacker can recover it. An audit log of the deletion must be preserved.

Breach Notification Requirements

Any data breach incident must be reported to the cyber authority within 48 hours of finding it. The affected users must be informed without any delay if their data is at risk. The notification email must include the nature of breach, what encrypted data was stolen, and mitigation steps. Organization must maintain a incident response plan for handling data breaches. After the incident, the company should maintain a record of the steps taken to handle it for future reference.

PART 6: Corporate Responsibilities

All companies and organizations operating in VIT's must comply with the following cybersecurity responsibilities. Create at least 10 specific, enforceable corporate obligations.

Required Responsibilities (Minimum 10)

Responsibility 1: Maintain Security Controls All companies and organization must deploy strong firewalls, IDS, IPS, and updated antivirus software to protect their encrypted databases from harmful network attacks. **Applies to:** All organizations and their IT departments. **Consequences of noncompliance:** A penalty fine of ₹5 lakh and mandatory government security audit.

Responsibility 2: Conduct Regular Security Audits Companies should conduct a deep cyber security audit of their encryption standards and server safety in every 6 months to find weaknesses. **Applies to:** IT administrators and compliance team. **Consequences of noncompliance:** Heavy financial fine and temporary suspension of digital operations.

Responsibility 3: Train Employees on Cybersecurity Every employee should be trained about phishing prevention, password security, and how to handle encrypted data safely every three months. **Applies to:** All employees and HR department. **Consequences of noncompliance:** Official warning to the company and mandatory retraining for the whole staff.

Responsibility 4: Encrypt Sensitive Customer Data Organization must protect customer's private and sensitive data by strictly encrypting it using AES-256 standard while storing it or sharing it across networks. **Applies to:** Data handlers and database administrators. **Consequences of noncompliance:** Immediate suspension of business license and fine up to ₹10 lakhs.

Responsibility 5: Maintain Incident Response Plan Organization should regularly test and update their incident response plan so they know exactly what actions to take during sudden data breaches or cyberattacks. **Applies to:** Cybersecurity team and senior management. **Consequences of noncompliance:** Fine up to ₹10 lakhs if the plan is found missing during an actual breach.

Responsibility 6: Data Breach Reporting Organization must report any data breach or encryption failure to the designated National Cyber Security Authority within 48 hours of finding it. **Applies to:** Compliance officer and top management. **Consequences of noncompliance:** For delayed reporting, the fine will increase up to ₹25 lakhs.

Responsibility 7: Implement Access Control Policies Access to sensitive or personal data should be given strictly on the basis of least privilege, meaning employees can only unlock data they need for their specific job. **Applies to:** IT administrator and regular employees. **Consequences of noncompliance:** Actions on negligent management and fine of ₹2 lakh.

Responsibility 8: Require Multi-Factor Authentication Multi factor authentication (MFA) should be implemented for all employees and system administrators for secure access to the company network. **Applies to:** All employees and administrators. **Consequences of noncompliance:** Fine of ₹1 lakh and a strict legal order to implement MFA immediately.

Responsibility 9: Secure Software Development While developing any new software or app, organization must use safe coding practices and run multiple penetration test before launching it to the public. **Applies to:** Software developers and QA testers. **Consequences of noncompliance:** If any damage happen due to insecure code, fine of ₹5 lakhs and free repair must be provided to users.

Responsibility 10: Disaster Recovery and Backups Organization should have a strong backup plan with offline servers to recover encrypted data fast if a ransomware attack happen. **Applies to:** System management and IT team. **Consequences of noncompliance:** Fine of ₹15 lakhs for losing customer data permanently due to negligence.

Examples to Consider

Maintain security controls | Conduct regular security audits | Train employees on cybersecurity | Report incidents within 72 hours | Protect customer information | Appoint a Data Protection Officer | Conduct annual penetration testing | Implement access control policies | Encrypt sensitive data | Maintain incident response plans

PART 7: Government Responsibilities

Specify how government agencies of VIT's must protect national interests in cyberspace. Include concrete mechanisms for each area.

Protect National Infrastructure

The government will protect critical infrastructure like central banks, city hospitals, power grid plants, and public transport systems. Because these systems are very important for the country, they must use military-grade encryption for all their databases. The government will also mandate strict air-gapping for physical control systems so hackers cannot access them through the public internet.

Monitor Cyber Threats

Government agencies will use advanced threat intelligence systems and AI surveillance tools to detect suspicious network activities in real time. They will set up a central monitoring protocol that constantly scan for zero-day exploits and ransomware attacks trying to bypass the national encryption standards. This will help the government stop the attacks before any data is stolen.

Investigate Cybercrimes

Special cybercrime units and digital forensic lab teams will be formed to investigate cybercrimes properly. To start an investigation, the police must get a legal digital warrant from the court first. After that, forensic experts will carefully extract the encrypted logs and electronic evidence without tampering it, so it is perfectly valid to be used in the court against the criminals.

International Coordination

Because cyber attacks can come from anywhere in the world, VIT's will heavily cooperate with other countries and international organizations like INTERPOL. They will sign information sharing agreements to exchange threat intelligence data quickly. If a foreign hacker attack VIT's infrastructure, the government will use international extradition treaties to bring them here for criminal trial.

Awareness Programs

The government will launch nationwide cybersecurity awareness campaigns in schools, colleges, and corporate companies. These educational programs will teach citizens how to use encrypted messaging apps, how to identify fake phishing links, and the importance of using strong passwords. They will also run video ads on social media platforms to educate regular people about their new data privacy rights.

PART 8: Cyber Incident Reporting Framework

Design a step-by-step process for how cyber incidents are reported, investigated, and resolved in VIT's. You must also create a flowchart (as a separate deliverable).

Reporting Process Steps

Step	Stage	Description (Student to Complete)
1	Incident Detected	The internal IT security team of the organization first notice unusual network traffic or a encryption failure. They have the responsibility to flag this to management immediately within 2 hours of detection.
2	Internal Investigation	The local cybersecurity team quickly investigate the server logs to find out how the hackers get in. They check if the AES-256 encryption lock is still safe or compromised. This must be done within 12 hours.
3	Report to Authority	The top management of the company must submit a official incident report to the National Cyber Security Authority (NCSA). By law, they have a strict timeline of 48 hours maximum to do this step.
4	Risk Assessment	NCSA officials and company experts analyze the overall impact. They check the severity of the attack. If the stolen database was fully encrypted, the risk to public is marked as low. This take about 72 hours.
5	Mitigation	The network administrators isolate the infected servers from the main internet. They stop the active attack and block the hackers from downloading any more encrypted files from the system.
6	Recovery	The IT department start restoring the digital services using secure offline backups. They also patch the software weakness so it don't happen again. Timeline for full recovery is usually 1 to 2 weeks.
7	Final Report	The company compliance officer submit a final detailed report to the NCSA. This document explain all the actions taken and lessons learned. It must be submitted within 30 days of the incident.

Deliverable Reminder

You must submit a separate FLOWCHART showing this complete process. Use tools like Canva, draw.io, Lucidchart, or PowerPoint SmartArt to create a professional diagram.

PART 9: National Cyber Security Authority (NCSA)

Design the organizational structure, responsibilities, and powers of the NCSA — the primary government agency responsible for cybersecurity in VIT's.

Organizational Structure

The NCSA will be headed by a Director General who report directly to the central government of VIT's. Under the Director General, there will be five Deputy Directors who manage the different specialized departments. All the local state-level cyber police stations will report directly to the NCSA headquarters to keep the communication fast during a cyber attack.

Core Responsibilities

The primary function of NCSA is to enforce the Comprehensive Data Encryption Act across the whole country. They are responsible for making new cybersecurity policies, monitoring national threats, and making sure all companies are encrypting their data properly. They also conduct public education campaigns to teach regular citizens about their digital privacy rights

Legal Powers

The NCSA have very strong legal power to protect the digital borders. They can legally force any private company or hospital to share their security audit reports and data logs if a national threat is detected. If a server or website is found spreading malware or hosting stolen encrypted data, the NCSA have the power to shut it down immediately without waiting for a court order.

Investigation Process

When a cybercrime complaint is received from a citizen, the NCSA first freeze the affected digital accounts to stop more damage. Then, the forensic team extract the server logs to find out the hackers IP address and how they bypassed the system. After the digital evidence is safely collected and verified, they identify the suspects and submit a final charge sheet to the cyber court for criminal trial.

Departments

To handle different cyber problems efficiently, the NCSA will be divided into 5 main specialized departments. These are the Threat Intelligence Unit, the Digital Forensics Lab, the Rapid Incident Response Team, the Cybercrime Investigation Unit, and the Public Awareness Division.

PART 10: AI and Emerging Technology Regulations

Create specific cybersecurity regulations for emerging technologies used in VIT's. For each technology, identify key risks and write enforceable rules.

Artificial Intelligence

Key Risks AI companies might use unencrypted citizen data to train their models without taking permission. Hackers can also use advanced AI tools to break weak encryption algorithms much faster than normal computers.

Regulations Any AI developer in VIT's must ensure their training datasets are fully encrypted at rest. They are strictly prohibited from using personal data to train models without taking explicit user consent first.

Deepfakes

Key Risks Criminals can use deepfake video or audio to bypass biometric security systems (like voice unlock for banks) or spread dangerous misinformation about VIT's political leaders.

Regulations All synthetic media and deepfakes generated in VIT's must contain a strong cryptographic watermark that proves it is AI-generated. Using deepfakes for identity theft or bypassing security is a strict non-bailable cybercrime.

IoT Devices

Key Risks Smart home devices (like security cameras) and smart city sensors usually comes with very weak default passwords. They also transmit user data over the network without proper encryption, making them easy target for hackers.

Regulations It is illegal for companies to sell IoT devices in VIT's that have hardcoded default passwords. Every smart device must mandate the user to create a strong password on setup, and all data sent to the internet must be encrypted using TLS 1.3 protocol.

Cloud Computing

Key Risks Because many companies store their data on the same public cloud server, a single data breach in the cloud infrastructure can expose millions of unencrypted files of different organizations at once.

Regulations Cloud service providers must enforce a strict zero-trust architecture. The database of each tenant (company) must be isolated and encrypted using their own separate cryptographic keys, so a breach in one account doesn't affect the others.

Digital Identity Systems

Key Risks If the central identity database of VIT's get hacked, the biometric and personal details of millions of citizens can be stolen permanently, leading to massive scale identity theft.

Regulations Biometric data and digital identity records must be cryptographically hashed and stored securely. The government must move towards decentralized identity systems so there is no single central server that hackers can target.]

Autonomous Systems

Key Risks Self-driving cars or delivery drones can be hacked remotely if the communication signals between the vehicle and the main server are not encrypted. Hackers can take physical control of the vehicle and cause accidents.

Regulations All remote control signals and navigation data for autonomous vehicles must be secured with military-grade encryption. The vehicles must also have a secure offline backup mode that activate immediately if the encrypted network connection is lost.

PART 11: Critical Infrastructure Protection

Create sector-specific cybersecurity requirements for each of the following critical infrastructure types in VIT's. Requirements should be practical and technically grounded.

Sector	Cybersecurity Requirements (Student to Complete)
Banks & Financial	• All customer transactions must use end-to-end encryption. • User passwords and ATM PINs should be cryptographically hashed. • Mandatory multi-factor authentication for all bank employees. • Secure offline backup of the daily financial ledgers. • Real-time AI monitoring to detect unusual encrypted traffic.
Hospitals & Healthcare	• Electronic Health Records (EHR) must be encrypted with AES-256. • Strict access control so only the assigned doctor can see data. • Regular offline backup to protect against ransomware attack. • Medical IoT devices (like smart monitors) must not have default passwords. • Encrypted communication channel between hospital branches.
Airports & Aviation	Flight control systems must be air-gapped from public Wi-Fi. • Radar and radio communication signals should be encrypted.

	• Strict biometric access control for the airport server rooms. • Regular penetration testing for the passenger ticketing system. • Emergency incident response plan to ground flights if hacked.
Railways & Transport	Train signaling networks must be encrypted to stop remote hacks. • Citizen ticket booking data must be protected using TLS 1.3. • CCTV camera footage must be encrypted during network transfer. • Mandatory cyber hygiene training for the control room staff. • Automated backup system for the live train schedules.
Power Plants & Energy	SCADA control system must be completely disconnected from internet. • Hardware-level encryption for the smart grid relays. • Zero-trust policy for any engineer doing remote maintenance. • Continuous monitoring to find zero-day exploits in smart meters. • Fast shutdown protocol if a cyber breach is detected in the grid.
Government Data Centers	Multi-region encrypted cloud storage to prevent data loss. • Localized cryptographic keys managed only by the NCSA.

- 24/7 physical and digital surveillance of the server clusters.
- Advanced firewall and DDOS protection must be active always.
- Routine security audits conducted by authorized ethical hackers.

PART 12: Awareness and Education

Design a National Cybersecurity Awareness Campaign for VIT's. Your campaign must reach all segments of society through age-appropriate and effective programs.

School Programs (Age 6-18)

For young students, the curriculum should teach basic digital safety like how to create strong passwords and how to identify a secure HTTPS website. High school students will do practical activities on how to use encrypted messaging apps and what to do if they face cyberbullying. The main learning outcome is to make them responsible digital citizens from a young age.

College & University Programs

Engineering students, specially in branches like EXCS or computer science, should have mandatory modules on using Python for network security and Machine Learning to detect encryption failures. Colleges must setup dedicated digital forensics lab so students can practice breaking and fixing encryption algorithms to earn their cyber certifications.

Corporate Training Programs

Every company must mandate their employees to take a cyber safety assessment every six months. The training will focus on teaching them how to safely handle and store encrypted customer data. They will also run fake phishing email simulations to test if the employees click on dangerous links.

Social Media Campaigns

The NCSA will run short video ads and infographics on platforms like Instagram and YouTube. The target demographic is mostly teenagers and older people who are easy targets for scammers. The messaging themes will be simple, like why they should never share their bank OTP or how to lock their social media profiles.

Public Awareness Workshops

Community workshops will be held every month in local town halls or public libraries. The target audience is local small business owners and parents. The delivery method will be live demonstration by cyber experts on how to install good antivirus software and how to backup their shop data securely.

PART 13: International Cooperation

Define how VIT's will collaborate with other countries and international organizations to combat cybercrime that crosses national borders.

Cybercrime Investigations

VIT's will form a global task force following frameworks similar to the Budapest Convention. When a foreign hacker steal our encrypted data, our NCSA will work directly with INTERPOL and foreign police to track the attacker's international IP address and gather digital evidence legally.

Information Sharing Agreements

VIT's will sign mutual legal assistance treaties (MLAT) with friendly nations. This agreement allow both countries to share threat intelligence and encryption vulnerability reports very fast without waiting for long political approvals, which is very important during an active cyber attack.

Joint Operations

If a big ransomware syndicate attack multiple countries at the same time, VIT's will actively participate in joint cyber operations. Our NCSA incident response team will team up with international agencies to raid the hacker's physical location and take down their server network together.

Extradition Requests

Under the new encryption act, any foreign criminal who bypass VIT's national infrastructure will face strict extradition. We will use international treaties like the Budapest Convention to formally request other countries to arrest the suspect and bring them to VIT's cyber court for their final criminal trial.

Threat Intelligence Exchange

VIT's will setup a real-time threat intelligence portal. International cyber agencies can upload data about new zero-day exploits or malware signatures here. This help our automated systems to update the national firewalls and block the new viruses before they even reach our country's network.

PART 14: Case Studies

Create 3 original fictional cybercrime cases set in VIT's. Each case must be realistic, detailed, and demonstrate how your law would be applied.

Case Study 1: The Wadala University Server Breach

Crime Category	Unauthorized Access / Data Breach
What Happened	In November 2025, a group of bad hackers bypassed the main firewall of Wadala Engineering College in VIT's. They successfully downloaded the personal records of 5000 students. But because the college followed the new encryption law and stored everything using strict AES-256 encryption, the hackers only got unreadable secret code files and couldn't read any real data.
Law Violated	Part 4: Unauthorized Access and Data Theft.
Investigation Process	The NCSA digital forensic lab checked the college server logs and found a zero-day exploit in the student portal. They tracked the attacker's IP address to a local cyber cafe and arrested two suspects.
Punishment Given	5 years imprisonment and ₹5 lakh fine for the hackers
Lessons Learned	This case proved that even if network security fail, the mandatory data encryption rule successfully saved the students privacy from being leaked.

Case Study 2: The Fake Nashik Transport Portal

Crime Category	Financial Fraud / Phishing
What Happened	In January 2026, scammers created a fake driving license and RTO appointment website for Nashik city in VIT's. They sent phishing SMS to citizens. When people tried to pay the exam fees, the website captured their credit card details in plaintext because the fake site didn't use proper TLS encryption.
Law Violated	Part 4: Phishing and Financial Fraud.
Investigation Process	The NCSA threat intelligence unit received multiple reports from citizens. Under their legal powers, they shut down the fake website immediately. They traced the unencrypted payment gateway logs to a local scammer group and arrested them.
Punishment Given	7 years imprisonment and ₹15 lakh fine.

Lessons Learned	The government need to run more public awareness programs so citizens know how to check for HTTPS and secure encrypted connections before making online payments.
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Case Study 3: The Smart Hospital Ransomware Attack

Crime Category	Ransomware / Cyber Terrorism
What Happened	A foreign cyber terrorist group injected a deadly ransomware into a central hospital in VIT's using a phishing email sent to a nurse. The malware tried to lock the Electronic Health Records (EHR). However, because the hospital had strict access control and offline encrypted backups, the terrorists couldn't lock the main server.
Law Violated	Part 4: Ransomware and Cyber Terrorism.
Investigation Process	The hospital IT team reported the incident to NCSA within 48 hours. NCSA isolated the network and worked with INTERPOL to track the international hackers using their threat intelligence exchange portal.
Punishment Given	Life imprisonment and ₹50 lakh fine for the caught terrorists.
Lessons Learned	Having strict offline backups and multi-factor authentication for hospital staff is very necessary to stop ransomware from spreading in critical national infrastructure.

PART 15: Future Recommendations

Based on your research, suggest forward-looking improvements and additions to VIT's cybersecurity law for the next 10 years. Consider how technology will evolve and what new threats will emerge.

AI Security (2025-2035)

As AI become more powerful, hackers will use advanced Machine Learning models to write dangerous malware and try to bypass our encryption. VIT's must regulate AI by making it mandatory to have a human review for any autonomous AI decision-making in critical sectors. Also, all companies must train their AI models on fully encrypted datasets to prevent data leaks and reduce AI bias.

Quantum Computing

Quantum computers are getting very fast and in the near future, they can easily break our current AES-256 encryption algorithms. VIT's must start preparing for this quantum threat right now. The government should mandate all critical organizations, especially banks and healthcare, to upgrade to Post-Quantum Cryptography (PQC) by the year 2030 so our encrypted data remain safe.

Privacy Protection

Because smart home devices and mobile apps collect so much background data today, our privacy laws need to be much stronger in the future. Biometric data like fingerprints and face scans should never be stored by companies unless they are cryptographically hashed. The law must also strictly ban behavioral profiling by social media platforms without taking explicit user permission first.

Digital Identity

VIT's should build a new national digital identity framework that is decentralized. Centralized identity systems are very risky because if that one main server get hacked, everyone's identity is stolen at once. By using decentralized hashed encryption, citizens can prove their identity securely without actually sharing their private data with third-party websites.

Cyber Warfare

In the next 10 years, state-sponsored cyberattacks from other countries will increase. If a foreign government try to hack and shut down VIT's critical infrastructure like power grids or transport systems, the law should officially define it as an act of cyber war. VIT's response policy must include active digital defense and immediately blocking all network traffic from the attacking country.

Evaluation Criteria

Your project will be graded based on the following criteria. Review this carefully to maximize your marks.

Done	Deliverable	Requirements
☐	Project Report	Minimum 25 pages, wellformatted, all 15 parts completed, diagrams included
☐	PowerPoint Presentation	15 to 20 slides, covering all major sections of the law
☐	Flowchart: Cybercrime Reporting	Shows complete incident reporting process from detection to resolution

[]	Flowchart: Investigation Process	Shows NCSA investigation steps from complaint to judgment
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Criteria	Max Marks	Marks Obtained
Research Quality — depth and accuracy of information, use of real-world examples	20	
Law Structure — completeness, logical organization, and legal clarity of the law	20	
Innovation — originality of solutions, creative approach to cybersecurity challenges	15	
Technical Accuracy — correctness of technical details, cybersecurity concepts, and procedures	15	
Presentation — quality of the PowerPoint presentation and oral delivery	10	
Diagrams & Flowcharts — clarity, accuracy, and professionalism of visual deliverables	10	
Case Studies — realism, detail, and correct application of the law	10	
TOTAL	100	

Deliverables Checklist

Before submitting, ensure you have completed all required deliverables:

Done	Deliverable	Requirements
[]	Flowchart: Data Breach Response	Shows data breach response procedure for organizations

[]	Infographic: Citizens' Rights	Visual summary of all citizens' digital rights
[]	Infographic: Cybercrime Categories	Visual taxonomy of all defined cybercrimes with penalties
[]	Infographic: Security Framework	Visual overview of the data protection and corporate security framework

Good Luck!

This project challenges you to think like a lawmaker, a technologist, and a citizen. Research realworld cybersecurity laws (India's IT Act, GDPR, Budapest Convention) for inspiration. Be creative, be thorough, and make VIT's digital future secure!